

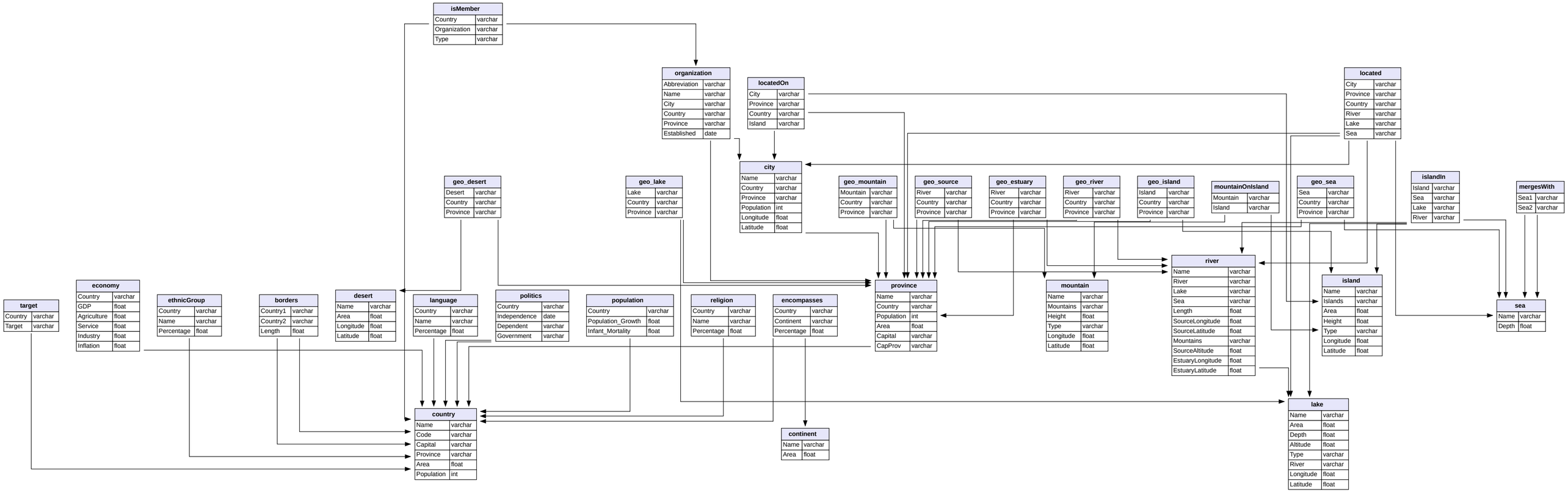
Università degli Studi di Salerno

Mondial: profili statistici dei Paesi del Mondo

Laboratorio di Informatica
Statistica per i Big Data

Anno Accademico
2023/2024

Mondial


isMember
Country varchar
Organization varchar
Type varchar
organization
Abbreviation varchar
Name varchar
City varchar
Country varchar
Province varchar
Established date
locatedOn
City varchar
Province varchar
Country varchar
Island varchar
located
City varchar
Province varchar
Country varchar
River varchar
Lake varchar
Sea varchar
city
Name varchar
Country varchar
Province varchar
Population int
Longitude float
Latitude float
geo_desert
Desert varchar
Country varchar
Province varchar
geo_lake
Lake varchar
Country varchar
Province varchar
geo_mountain
Mountain varchar
Country varchar
Province varchar
geo_source
River varchar
Country varchar
Province varchar
geo_estuary
River varchar
Country varchar
Province varchar
geo_river
River varchar
Country varchar
Province varchar
geo_island
Island varchar
Country varchar
Province varchar
mountainOnIsland
Mountain varchar
Island varchar
geo_sea
Sea varchar
Country varchar
Province varchar
islandIn
Island varchar
Sea varchar
Lake varchar
River varchar
mergesWith
Sea1 varchar
Sea2 varchar
target
Country varchar
Target varchar
economy
Country varchar
GDP float
Agriculture float
Service float
Industry float
Inflation float
ethnicGroup
Country varchar
Name varchar
Percentage float
borders
Country1 varchar
Country2 varchar
Length float
desert
Name varchar
Area float
Longitude float
Latitude float
language
Country varchar
Name varchar
Percentage float
politics
Country varchar
Independence date
Dependent varchar
Government varchar
population
Country varchar
Population_Growth float
Infant_Mortality float
religion
Country varchar
Name varchar
Percentage float
encompasses
Country varchar
Continent varchar
Percentage float
province
Name varchar
Country varchar
Population int
Area float
Capital varchar
CapProv varchar
mountain
Name varchar
Mountains varchar
Height float
Type varchar
Longitude float
Latitude float
river
Name varchar
River varchar
Lake varchar
Sea varchar
Length float
SourceLongitude float
SourceLatitude float
Mountains varchar
SourceAltitude float
EstuaryLongitude float
EstuaryLatitude float
island
Name varchar
Islands varchar
Area float
Height float
Type varchar
Longitude float
Latitude float
sea
Name varchar
Depth float
country
Name varchar
Code varchar
Capital varchar
Province varchar
Area float
Population int
continent
Name varchar
Area float
lake
Name varchar
Area float
Depth float
Altitude float
Type varchar
River varchar
Longitude float
Latitude float

Focus dell’analisi

Focus 1

Esiste una relazione tra il tasso di crescita della popolazione e il tasso di mortalità infantile?

Focus 2

Quanto è forte il legame tra il PIL e la popolazione?

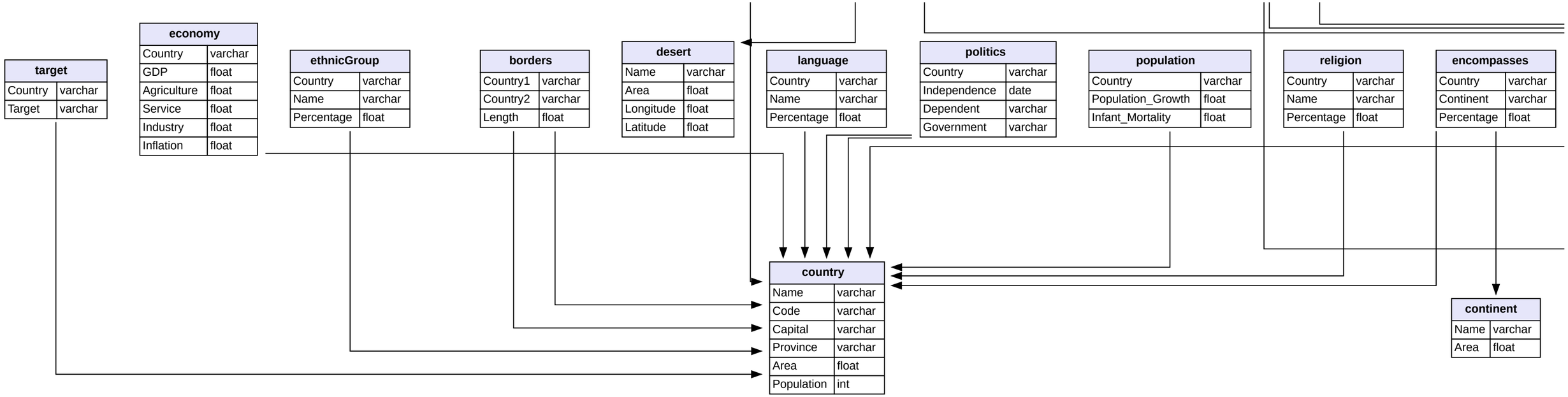
Focus 3

Che tipo di associazione sussiste tra la religione predominante e la forma di governo in un Paese?

Focus 4

La religione predominante professata nei Paesi dipendenti coincide con quella professata nei Paesi da cui dipendono?

Mondial: zoom



	Continent	Nation	Code	Population	Government	Dependent	GDP	Infant_Mortality	Population_Growth	Religion
0	Europe	Austria	A	8023244	federal republic	None	152000.0	6.20	0.41	Roman Catholic
1	Asia	Afghanistan	AFG	22664136	transitional government	None	12800.0	149.70	4.78	Muslim
2	Europe	Albania	AL	3249136	emerging democracy	None	4100.0	49.20	1.34	Muslim
3	Australia/Oceania	American Samoa	AMSA	65628	unincorporated and unorganized territory of th...	USA	462.2	10.18	1.22	Christian Congregationalist
4	Europe	Andorra	AND	72766	parliamentary democracy that retains as its he...	None	1000.0	2.20	2.96	Roman Catholic



Importiamo i dati:
MySQL

```sql
SELECT encompasses.Continent, country.Name AS Nation, country.Code, country.Population, politics.Government,
       politics.Dependent, economy.GDP, population.Infant_Mortality, population.Population_Growth,
       religion.Name AS Religion
FROM country
JOIN encompasses ON country.Code = encompasses.Country
JOIN economy ON country.Code = economy.Country
JOIN population ON country.Code = population.Country
JOIN politics ON country.Code = politics.Country
JOIN religion ON country.Code = religion.Country
WHERE religion.Percentage = (
    SELECT MAX(r2.Percentage)
    FROM religion r2
    WHERE r2.Country = country.Code
) AND encompasses.Percentage = (
    SELECT MAX(e2.Percentage)
    FROM encompasses e2
    WHERE e2.Country = country.Code);
```



Importiamo i dati: Python

Librerie utilizzate: *mysqlalchemy*, *pandas*

```
pip install mysql-connector-python
```

```
engine = create_engine("mysql+mysqlconnector://guest:relational@db.relational-data.org/Mondial", echo=False)
```

```
c = engine.connect()
```

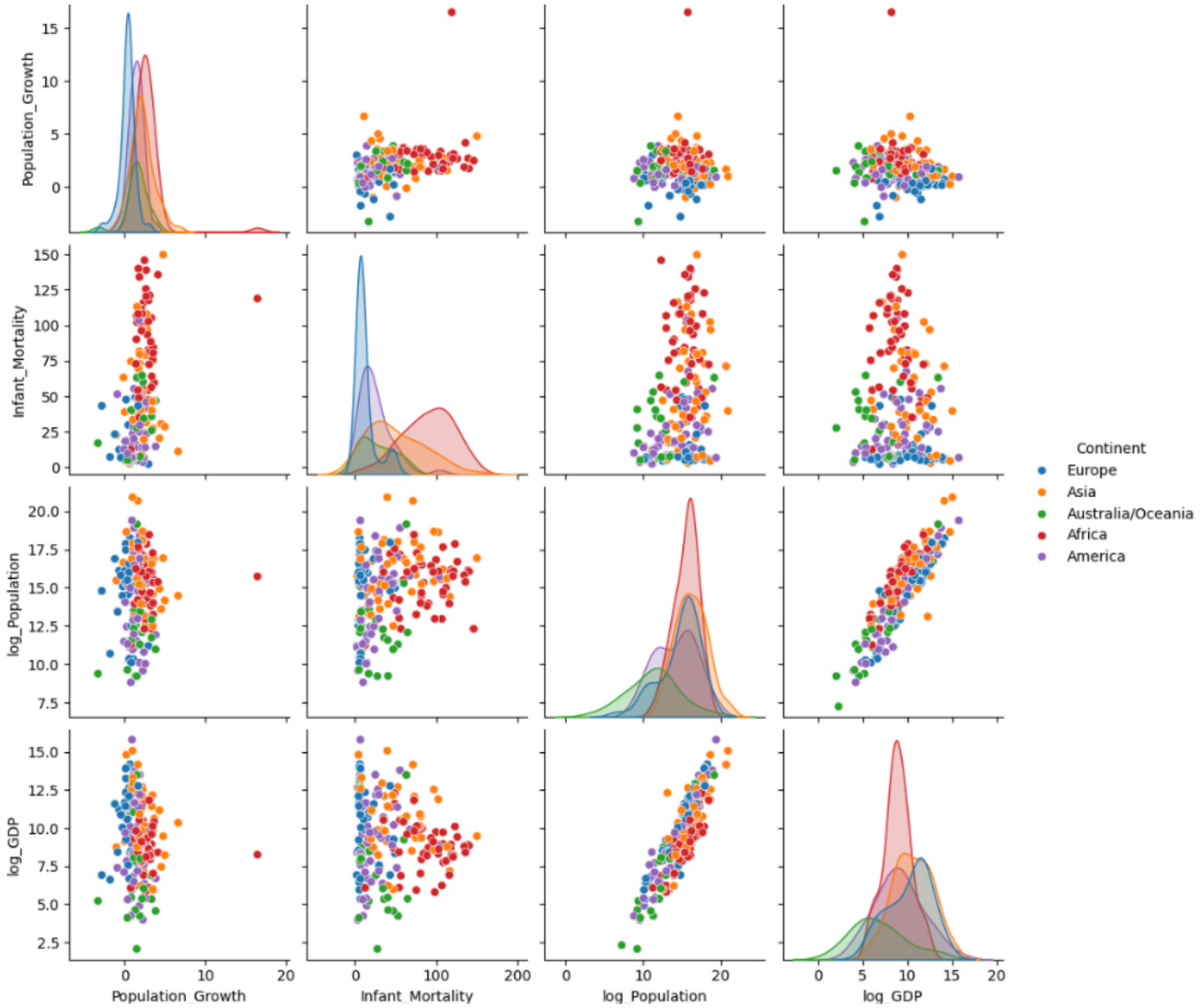
```
q = """SELECT encompasses.Continent, country.Name AS Nation, country.Code,
       country.Population, politics.Government, politics.Dependent, economy.GDP, population.Infant_Mortality,
       population.Population_Growth, religion.Name AS Religion
FROM country
JOIN encompasses ON country.Code = encompasses.Country
JOIN economy ON country.Code = economy.Country
JOIN population ON country.Code = population.Country
JOIN politics ON country.Code = politics.Country
JOIN religion ON country.Code = religion.Country
WHERE religion.Percentage = (
    SELECT MAX(r2.Percentage)
    FROM religion AS r2
    WHERE r2.Country = country.Code
) AND encompasses.Percentage = (
    SELECT MAX(e2.Percentage)
    FROM encompasses AS e2
    WHERE e2.Country = country.Code)
"""
```

```
data = pd.read_sql_query(text(q), c)
```

Analisi esplorativa: summary

	Population	GDP	Infant_Mortality	Population_Growth
count	2.090000e+02	2.030000e+02	202.000000	203.000000
mean	2.561049e+07	1.551136e+05	45.519208	1.732005
std	1.098337e+08	6.234617e+05	39.699974	1.691520
min	4.800000e+01	7.800000e+00	2.200000	-3.300000
25%	4.276420e+05	2.500000e+03	9.687500	0.790000
50%	4.394537e+06	1.040000e+04	33.650000	1.710000
75%	1.426156e+07	7.130000e+04	75.100000	2.495000
max	1.210005e+09	7.247700e+06	149.700000	16.490000

Analisi esplorativa: pairplot



Librerie utilizzate: *seaborn*, *numpy*

Le variabili log_GDP e log_Population sono state ottenute tramite le trasformazioni logaritmiche delle variabili GDP e Population.

Focus 1

Si analizza la relazione tra il tasso di crescita della popolazione e il tasso di mortalità infantile.

```
q95_pop_growth = data1['Population_Growth'].quantile(0.95)
q95_inf_mort = data1['Infant_Mortality'].quantile(0.95)

data_q95 = data1[(data1['Population_Growth'] <= q95_pop_growth) &
                 (data1['Infant_Mortality'] <= q95_inf_mort)]

sns.lmplot(x = "Population_Growth", y = "Infant_Mortality", data = data_q95, fit_reg = True)
```

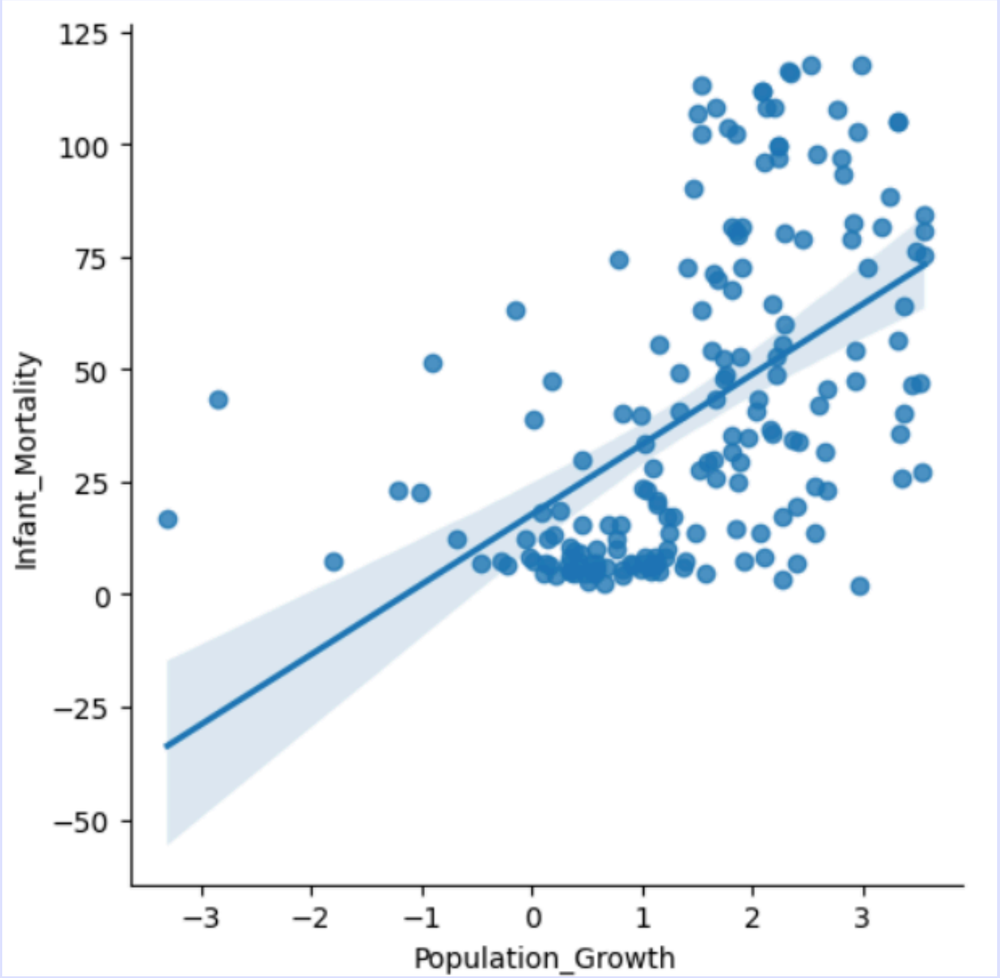
N.B. data1 è il dataset ottenuto rimuovendo le osservazioni nulle dalle colonne interessate

```
lr1 = stats.linregress(data_q95["Population_Growth"], data_q95["Infant_Mortality"])
```

```
lr1.rvalue
```

```
0.5234704962579145
```

Librerie utilizzate:
seaborn, scipy



Classificatore per i continenti

Librerie utilizzate: *sklearn*

Sulla base della relazione precedente, si costruisce un classificatore bayesiano per identificare il continente di appartenenza delle singole nazioni.

```
classifier = GaussianNB()
```

```
X = data1[['Infant_Mortality', 'Population_Growth']]
Y = data1["Continent"]
data1_cl = pd.concat([X, Y], axis = 1)
```

```
x_train, x_test, y_train, y_test = train_test_split(data1_cl[data1_cl.columns[:2]], data1_cl["Continent"], test_size = 0.2, random_state=99)
```

```
classifier.fit(X=x_train, y=y_train)
```

```
▾ GaussianNB
GaussianNB()
```

```
pred = classifier.predict(x_test)
```

```
accuracy_score(y_test, pred)
```

```
0.7317073170731707
```

Focus 2

Si analizza la relazione tra il PIL e la popolazione.

```
sns.lmplot(x = "log_Population", y = "log_GDP", data = data2, fit_reg = True)
```

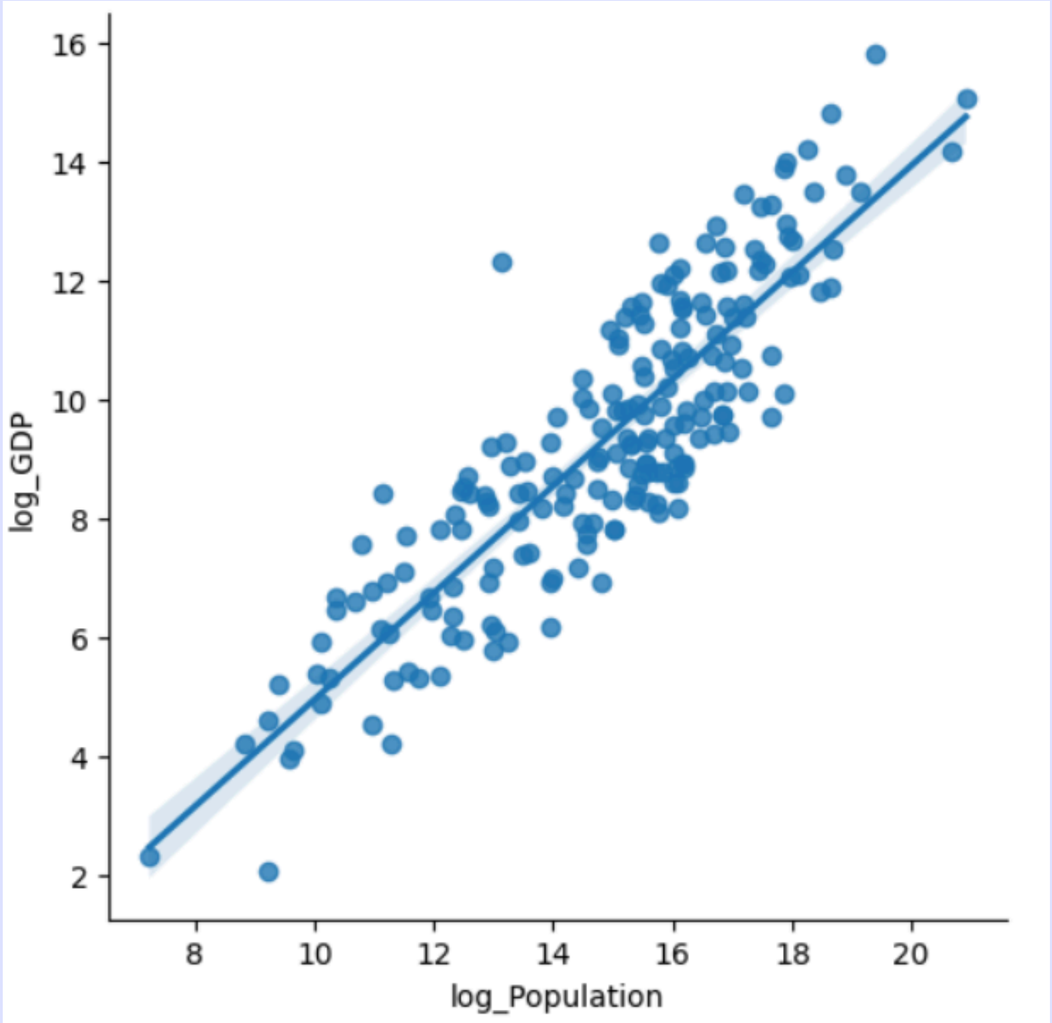
N.B. data2 è il dataset ottenuto rimuovendo le osservazioni nulle dalle colonne interessate

```
lr2 = stats.linregress(data2["log_GDP"], data2["log_Population"])
```

```
lr2.rvalue
```

```
0.8730311523051977
```

Librerie utilizzate:
seaborn, scipy



Focus 3

Si individua la religione più presente nei paesi che sono organizzati con una delle cinque forme di governo piú diffuse.

```
data_gov_rel = data.groupby(['Government', 'Religion']).size().reset_index(name='Count')

max_indices = data_gov_rel.groupby('Government')['Count'].idxmax()

rel_per_gov = {}

for idx in max_indices:
    row = data_gov_rel.loc[idx]
    government = row['Government']
    religion = row['Religion']
    rel_per_gov[government] = religion

top5_gov = data["Government"].value_counts().head(5)

rel_top5_gov = {}

for gov, rel in rel_per_gov.items():
  if gov in top5_gov:
    rel_top5_gov[gov] = rel
```

```
{'British Overseas Territories': 'Protestant',
 'constitutional monarchy': 'Roman Catholic',
 'federal republic': 'Roman Catholic',
 'parliamentary democracy': 'Roman Catholic',
 'republic': 'Roman Catholic'}
```

PIL - Governo - Religione

Si selezionano i paesi che presentano una delle coppie governo-religione appena individuate.
Si studia il PIL di tali paesi in confronto al PIL dei restanti.

```
data[data['Nation'].isin(nation_sel)].groupby("Continent")["GDP"].describe()
```

Continent	count	mean	std	min	25%	50%	75%	max
Africa	7.0	6181.428571	7352.188855	430.0	1470.00	2500.0	10300.00	16800.0
America	21.0	96661.952381	216303.062498	53.0	6500.00	17000.0	87000.00	976800.0
Asia	2.0	90094.500000	126721.313363	489.0	45291.75	90094.5	134897.25	179700.0
Australia/Oceania	1.0	68.000000	NaN	68.0	68.00	68.0	68.00	68.0
Europe	16.0	218448.000000	366124.640533	380.0	17575.00	89350.0	168125.00	1173000.0

Paesi selezionati

```
data[~data['Nation'].isin(nation_sel)].groupby("Continent")["GDP"].describe()
```

Continent	count	mean	std	min	25%	50%	75%	max
Africa	47.0	24328.693617	4.518084e+04	325.0	3800.0	7400.0	20000.0	215000.0
America	22.0	401623.772727	1.542898e+06	68.0	1375.0	3825.0	14425.0	7247700.0
Asia	43.0	281757.906977	6.839498e+05	390.0	8200.0	41100.0	247300.0	3500000.0
Australia/Oceania	20.0	25011.660500	9.058533e+04	7.8	162.4	438.6	3543.5	405400.0
Europe	24.0	131019.291667	3.065067e+05	205.0	3550.0	32900.0	105825.0	1452200.0

Paesi restanti

```
nation_sel = []

for index, row in data.iterrows():
    governo = row["Government"]
    religione = row["Religion"]
    if (governo, religione) in rel_top5_gov.items():
        nation_sel.append(row["Nation"])
```

Focus 4

Data la religione predominante per i paesi non indipendenti, si verifica l'eventuale corrispondenza con la religione predominante dei rispettivi paesi da cui dipendono.

```
dep_nat = data[data['Dependent'].notna()]

indep_nat = data[data['Dependent'].isna()]

rel_df = dep_nat.merge(indep_nat, left_on = 'Dependent', right_on = 'Code')
```

```
rel_df['Match'] = rel_df['Religion_Dependent'] == rel_df['Religion_Independent']
```

```
rel_df.head()
```

	Nation	Dependent	Religion_Dependent	Religion_Independent	Match
0	American Samoa	USA	Christian Congregationalist	Protestant	False
1	Guam	USA	Roman Catholic	Protestant	False
2	Puerto Rico	USA	Roman Catholic	Protestant	False
3	Aruba	NL	Roman Catholic	Roman Catholic	True
4	Cocos Islands	AUS	Muslim	Anglican	False

Percentuale totale di corrispondenza tra le religioni

```
rel_df['Match'].sum() / len(rel_df) * 100
```

```
52.38095238095239
```

Grazie per l'attenzione!

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